Kostadin Rangelov Kostadinov

MD, PhD, MPH, MSc · Chief Assistant Professor, Department of Social Medicine and Public Health, Medical University of Plovdiv

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Current appointments

- **Chief Assistant Professor**, Department of Social Medicine and Public Health, Faculty of Public Health, Medical University of Plovdiv *(July 2025 – present)*
- **Founder and Chief Executive Officer**, Social Epidemiology Lab — independent research and analytics practice in epidemiology, health economics and health policy
- **Resident Physician in Infectious Disease Epidemiology**, Department of Epidemiology and Disaster Medicine, Medical University of Plovdiv *(November 2025 – present)*
- **Attending Physician**, Medicus Alpha Hospital, Plovdiv *(December 2020 – present)*

Researcher identifiers

ORCID 0000-0003-1414-8738 · Google Scholar · Scopus Author ID 57542360900 · Web of Science AEE-9300-2022 · GitHub · LinkedIn · X

Last updated: September 2026.

Academic profile

Physician-epidemiologist and Chief Assistant Professor in the Department of Social Medicine and Public Health at the Medical University of Plovdiv, working at the intersection of epidemiology, health economics and health policy. Research centres on how health systems allocate scarce resources — orphan medicines and rare cancers, the cost of illness, health technology assessment, and the health consequences of the urban environment — using routinely collected administrative health data and real-world evidence.

Holds an MD, a PhD in social medicine and public health, an MPH, and master's degrees in economics and finance, statistical business consulting, and impact assessment and policy appraisal, and is currently reading for a Master of Laws. Board-certified in Social Medicine and Organisation of Health Care and Pharmacy, and currently in residency training in Infectious Disease Epidemiology.

Develops and publishes open research software: `kkstatfun`, an R toolkit of 146 functions for medical statistics, epidemiology and health economics. Associate Editor at JMIR Publications and founder of the Social Epidemiology Lab.

Research interests

Epidemiology · Social medicine and public health · Infectious disease epidemiology · Health policy and health technology assessment · Real-world evidence and health economics · Environmental and urban health · Biostatistics and reproducible research

The publication record is organised around six research programmes:

1. **Rare diseases, rare cancers and access to orphan medicines.** Policy analysis of definitions, registration, organisation of care and reimbursement; timeliness of market access; national and EU-level policy alternatives.
2. **Health economics, HTA and real-world evidence.** Cost-of-illness studies (myasthenia gravis, spinal muscular atrophy, osteoporosis, end-stage kidney disease, terminal cancer), budget-impact validation, and public expenditure analysis from a payer perspective.
3. **Infectious disease epidemiology and antimicrobial resistance.** COVID-19 epidemiology and vaccine impact, urinary tract and sexually transmitted infections, antimicrobial resistance trends, echinococcosis, and HPV-attributable expenditure.
4. **Environmental and urban health.** Air pollution and cardiovascular risk, green space and land cover, health impact assessment in spatial planning, and stakeholder evidence for urban environmental policy.
5. **Health professions education and digital health.** Virtual-reality and simulation-based training, trauma-informed care, and mental health literacy among school and health professionals.
6. **Applied biostatistics, prediction and reproducible research.** External validation of clinical prediction and AI tools, diagnostic accuracy, and open, reproducible statistical software.

Academic appointments

July 2025 - Present

- **Chief Assistant Professor** | Department of Social Medicine and Public Health | Faculty of Public Health | Medical University of Plovdiv
  - Lectures and practical training in Medical Ethics, Biostatistics, Social Medicine and Public Health, Public Dental Health, and Clinical Trial Design for Non-Interventional Studies
  - Course coordination and curriculum development for undergraduate public health and biostatistics teaching
  - Design and marking of written and practical examinations; student assessment
  - Supervision and mentoring of student research projects
  - Research within the departmental programme in health policy, health economics and environmental health

September 2020 - July 2025

- **Assistant Professor** | Department of Social Medicine and Public Health | Faculty of Public Health | Medical University of Plovdiv
  - Practical classes in Medical Ethics, Biostatistics, Social Medicine and Public Health

Research appointments

October 2024 - July 2026

- **Junior Researcher (R1)** | Research Group 3.1.5 "Health and Quality of Life in a Green and Sustainable Environment" | Strategic Research and Innovation Programme for the Development of MU-Plovdiv (PSNIIR-MUP) | Contract BG-RRP-2.004-0007-C01 of 31 December 2022

2025 - Present

- **Founder and Chief Executive Officer** | Social Epidemiology Lab | Independent research and analytics practice: study design, epidemiological and statistical analysis, health economics and HTA, and policy impact assessment for universities, clinicians, NGOs and public institutions

Clinical appointments

December 2020 - Present

- **Attending Physician** | Medicus Alpha Hospital | Plovdiv, Bulgaria | General inpatient and outpatient practice across acute and chronic conditions and injuries

December 2017 - March 2020

- **Resident Physician in Cardiology** | Cardiology Clinic | University Hospital Plovdiv | Inpatient care

Academic education

2026 - Present

- **Master of Laws** | Burgas Free University | Enrolled; studies commence October 2026

October 2025 - July 2026

- **MSc in Impact Assessment and Policy Appraisal** | Department of Public Administration | Sofia University "St. Kliment Ohridski"

October 2024 - September 2025

- **MSc in Statistical Business Consulting** | D. A. Tsenov Academy of Economics, Svishtov

December 2021 - January 2025

- **PhD** | Department of Social Medicine and Public Health | Faculty of Public Health | Medical University of Plovdiv
  - **Dissertation:** *Health Policies in the Field of Rare Tumours*
  - **Field:** Medicine — Public Health — Social Medicine and Organisation of Health Care and Pharmacy
  - **Supervisors:** Prof. Georgi Iskrov, PhD and Prof. Nina Musurlieva, DMD, PhD
  - **Thesis submitted** 2024; **publicly defended** January 2025
  - Dissertation appendices and analysis code: github.com/kostadinoff/phd_appendices · PhD defence announcement: mu-plovdiv.bg

October 2021 - September 2022

- **MSc in Economics and Finance** | Faculty of Economics | Sofia University "St. Kliment Ohridski"

October 2017 - September 2019

- **MPH in Public Health and Health Management** | Faculty of Public Health | Medical University of Plovdiv

September 2012 - December 2017

- **MD (Master's in Medicine)** | Faculty of Medicine | Medical University of Plovdiv

Medical qualifications and specialty training

November 2025 - Present

- **Residency in Infectious Disease Epidemiology** | Department of Epidemiology and Disaster Medicine | Medical University of Plovdiv

April 2022 - May 2025

- **Board-certified specialist in Social Medicine and Organisation of Health Care and Pharmacy** | National medical specialty certification | Medical University of Plovdiv

October 2024 - July 2025

- **Postgraduate qualification in Sexology and Sexopathology** | Department of Postgraduate Education | Medical University of Plovdiv

Since 2017

- **Medical licensure** | Bulgarian Medical Association | Registration No. 1700005331

Research funding and projects

Grants, fellowships and personal funding

September 2026

- **Partnership for International Vaccine Initiatives (PIVI)** | The Task Force for Global Health | Consultancy contract with the Bulgarian Society for Innovative Medicine | Analysis of the burden of influenza, other acute respiratory infections and respiratory syncytial virus disease in Bulgaria | Role: **Consultant**

2026

- **World Health Organization** | Externally commissioned research contract | Epidemiological and economic analysis of varicella and varicella vaccination in Bulgaria, informing the introduction of varicella vaccine into the national immunisation programme in 2026 | Role: **Consultant — lead analyst**

September 2024 - 2025

- **National Science Programme "Young Scientists and Postdoctoral Researchers – 2"** | Ministry of Education and Science of the Republic of Bulgaria | Competitive personal funding | Role: **Awardee**

November 2025 - Present

- **Project BG05SFPR002-1.014-0001 "Encouraging specialisation in specialties and areas with a shortage of specialists"** | Human Resources Development Programme 2021–2027 | Role: **Beneficiary**

September 2023; January 2026

- **Erasmus+ Staff Mobility for Teaching grants** | University of West Attica, Greece (2023); Tartu Health Care College, Estonia (2026)

International projects

October 2026 - Present

- **1HEALTH-NET** — COST Action CA25101 "One Health Network for Infectious Diseases Detection, Monitoring and Prevention" | European Cooperation in Science and Technology (COST) | Role: **Working Group application** (under review by the Action)

2024 - 2026

- **AFFIRMO** — Atrial Fibrillation Integrated Approach in Frail, Multimorbid and Polymedicated Older People | European Union Horizon 2020 research and innovation programme | Grant Agreement No. 899871 | Role: **National site monitor** — monitoring of study conduct and protocol compliance at the Bulgarian site, source data verification and implementation support

February 2022 - January 2025

- **Safe4Child** — "Caring for Violent Children Safely in Child Psychiatric and Residential Units" | Erasmus+ Programme, Key Action 2 (KA220-HED) | Agreement No. 2021-1-FI01-KA220-HED-000032106 | Coordinator: Turku University of Applied Sciences, Finland | Partners: HAW Hamburg (Germany), University College Cork (Ireland), Medical University of Plovdiv (Bulgaria) | Total budget EUR 358,235; MU-Plovdiv allocation EUR 41,778
  - Role: **Principal Investigator for Bulgaria**, Department of Social Medicine and Public Health, Faculty of Public Health, MU-Plovdiv
  - Contribution: led the Bulgarian site; design, statistical analysis and reporting of the evaluation of the virtual-reality trauma-informed care training programme
  - Main outputs: pre–post evaluation study in *JMIR Medical Education* (2026); qualitative multi-country study in *Child Care in Practice* (2025); systematic review in *Journal of Child and Adolescent Trauma* (2023)

September 2021 - 2025

- **Screen4Care** — "Shortening the Path to Rare Disease Diagnosis Using Newborn Genetic Screening and Digital Technologies" | Innovative Medicines Initiative 2 Joint Undertaking | Grant Agreement No. 101034427
  - Role: **Researcher** — landscape analysis of European data sources amenable to machine learning; mapping of rare disease registries and databases against organisational, FAIR and legal principles
  - Main outputs: project protocol in *PLOS ONE* (2023); scoping review in *Frontiers in Public Health* (2023); landscape analysis in *Orphanet Journal of Rare Diseases* (2024); machine-learning readiness study in *Orphanet Journal of Rare Diseases* (2024)

2020 - August 2023

- **W@S / Well@School** — "Developing Multi-Professional Higher Education for Promoting Mental Health and Well-Being in Schools" | Erasmus+ Programme, Key Action 2 | Project Reference 2020-1-FI01-KA203-066521 | Coordinator: Turku University of Applied Sciences, Finland | MU-Plovdiv lead: Prof. Gergana Petrova, PhD
  - Role: **Researcher** — leading contribution to the development, methodology, analysis and reporting of the training evaluation
  - Main output: Korhonen J, Lahti M, **Kostadinov K**, et al. Fostering Mental Health Literacy Among Primary School Professionals. *Int J Environ Res Public Health.* 2025;22:435

2025 - Present

- **"Combined Effects of Exposure to Natural and Built Environment – Informing Healthy and Sustainable Urban Development"** | National Science Programme with EU funding (Recovery and Resilience Plan)
  - Role: **Researcher** — epidemiological study design, exposure–outcome modelling and statistical analysis of urban environmental exposures and population health outcomes
  - Main outputs: stakeholder survey in *Urban Science* (2026); health impact assessment framework presented at AESOP 2025; land cover and allergic rhinitis analysis presented at ISEE 2026

March - May 2020

- **"Interdisciplinarity, Multiculturalism and Patient Care in Non-Standard Situations in the Context of Didactic Activities in Medical and Health Sciences at Medical Simulation Centres"** | Erasmus+ Programme, Key Action 2 | Project No. 2019-1-PL01-KA203-065205 | Role: **Researcher**

2018 - 2019

- **AEGIS II** — ApoA-I Event Reducing in Ischemic Syndromes II | International multicentre clinical trial | Role: **Study Coordinator** (site level)

National projects and programmes

2026 - Present

- **"Association of Transport Noise with Mental Health and Well-Being in Residents of Sofia: Underlying Mechanisms, Health and Socio-Economic Impact"** | Bulgarian National Science Fund, 2026 call for fundamental research | Contract КП-06-ПН05/20 (BNSF-1.011-0133) | Principal Investigator: Assoc. Prof. Angel Dzhambov, MD, PhD, DSc, Medical University of Plovdiv | Role: **Research team member**

December 2025 - June 2026

- **Academic Mentor** | Project BG05SFPR001-3.002-0001 "From Higher Education to Employment" | Mentoring of students in practical placements; supporting the match between higher education provision and labour market demand

October 2024 - July 2026

- **Research Group 3.1.5 "Health and Quality of Life in a Green and Sustainable Environment"** | Strategic Research and Innovation Programme for the Development of MU-Plovdiv | Agreement BG-RRP-2.004-0007-C01 of 31 December 2022 | Role: **Junior Researcher (R1)**

University projects

2026

- **"Conflict Dynamics and Illness Behaviour"** | Project No. PHO1400 | Project Leader: Assoc. Prof. Zlatoslav Arabadzhiev, MD, PhD | Role: **Statistical analysis and methodology** | Medical University of Plovdiv

2025

- **"Investigation of New Modern Biomarkers for Diagnosis and Therapeutic Response in Urinary Tract Infections"** | PhD and Postdoctoral Project DPDP-11/2025 | Role: **Epidemiological design and statistical analysis** | Medical University of Plovdiv
- **"Healthcare Costs and Their Economic and Social Effects – Statistical Dimensions"** | Inter-university research project, nationally funded | Role: **Researcher — health economics and statistical analysis** | Main output: Kostadinov K, Raycheva R, Zdravkova-Aneva I, Shopova M, Ovchinnikov E, Petkov P. Accuracy of Budget Impact Projections in Bulgarian Health Technology Assessment. *Healthcare*. 2025;13:2990
- **"Development of a Scientific Framework and National Methodology for Psychotherapy of Dysfunctional Schemas Based on Evidence"** | Intra-university research project No. HO-2/2025 | Principal Investigator: Kristina Stoyanova, PhD (NIMH – Plovdiv) | Scientific Supervisor: Assoc. Prof. Zlatoslav Arabadzhiev, MD, PhD | Role: **Statistical analysis and methodology** | Medical University of Plovdiv

September 2023 - 2025

- **"Study of the Relationship Between Non-Cultivable and Hard-to-Culture Microorganisms and Fertility Function in Men with Symptomatic and Asymptomatic Infections of the Lower Urogenital Tract"** | University Project No. HO-17/2023 | Role: **Epidemiological design and statistical analysis** | Medical University of Plovdiv

October 2023

- **"Comparative Study on Modern Microbiological Methods for Rapid Etiological Diagnosis of Urinary Tract Infections"** | PhD and Postdoctoral Project | Agreement No. 10/2023 | Role: **Statistical analysis** | Medical University of Plovdiv

September 2022

- **"Antimicrobial Activity of Root Canal Fillers for the Treatment of Endodontic Infection in Primary Teeth"** | PhD and Postdoctoral Project | Agreement No. 04/01.09.2022 | Role: **Statistical analysis** | Medical University of Plovdiv

September 2021

- **"Rapid Microbiological Diagnosis of Genital Infections in Women and Men – Comparative Analysis"** | Project No. HO-03/2020 (HO-P-8445) | Funded by the Bulgarian Ministry of Education and Science under the regulations for targeted state budget allocations for scientific and creative activities in higher education institutions | Role: **Statistical analysis** | Medical University of Plovdiv

Publications

Peer-reviewed original research articles (46)

1. Petkov P, Shopova M, Ovchinnikov E, Raycheva R, **Kostadinov K**. Determinants of Government Health Expenditure in the European Union: Robust Panel Evidence from the EU-27, 2000–2023. *Economies*. 2026;14(10):453. doi:10.3390/economies14100453

2. Dzhambov AM, Kitov S, **Kostadinov KR**, Kitova MF, Burov A, Helbich M, Deneva T, Kitova L. Residential exposures associated with adiposity and proinflammatory markers in obese adults with metabolic syndrome: a cross-sectional study. *Archives of Industrial Hygiene and Toxicology*. 2026;77(3):198–212. doi:10.2478/aiht-2026-77-4124

3. Zdravkova-Aneva I, Dermendzhiev T, **Kostadinov K**, Prodanov A, Dimova R, Raycheva R. Physicians' Perspectives on Access to Innovative Oncology Therapies: A Cross-Sectional Survey in Bulgaria. *Healthcare*. 2026;14(18):2926. doi:10.3390/healthcare14182926

4. Gyokova E, Hristova-Atanasova E, **Kostadinov K**, Iskrov G, Andreeva A. Towards optimized preterm birth prediction: The combined value of cervical biometry and biochemical biomarkers. *Journal of IMAB - Annual Proceeding (Scientific Papers)*. 2026;32(3):7002–7012. doi:10.5272/jimab.2026323.7002

5. **Kostadinov K**, Mandova V, Rangelova V, Kevorkyan A, Raycheva R. Estimated HPV-Attributable Inpatient Pharmaceutical Reimbursement in Bulgaria, 2021–2025: A National Payer-Perspective Expenditure Analysis. *Healthcare*. 2026;14(16):2474. doi:10.3390/healthcare14162474

6. Lip GYH, Proietti M, Ainsworth J, Dan GA, Frost L, Graffigna G, Lane DA, Lucci D, Fabbri G, Marin F, O'Flaherty M, Petrovic M, Potpara TS, Proietti R, Sanaullah A, Tokmakova M, Vetrano DL, Johnsen SP, Maggioni AP, AFFIRMO Study Investigators and Consortium. Integrated care management and comprehensive geriatric assessment using a mHealth-based approach in older multimorbid patients with atrial fibrillation: the AFFIRMO cluster-randomised trial. *The Lancet*

Regional Health – Europe. 2026:101832. doi:10.1016/j.lanepe.2026.101832 **Group authorship: Kostadin Kostadinov (AFFIRMO Study Investigator and Consortium collaborator).**

7. Markova P, Yaneva A, Spasova M, Markov S, **Kostadinov K**. Assessment of NGAL, eGFR, and tubular injury markers for early detection of chemotherapy-induced nephrotoxicity in pediatric oncology patients. *Folia Medica*. 2026;68(4):e184913. doi:10.3897/folmed.68.e184913

8. Popova-Daskalova G, **Kostadinov K**, Eneva K, Masarlieva A, Vuchev D, Uchikov A, Aleksova L, Nikolov N, Nedkova T. Postoperative Prophylaxis of Cystic Echinococcosis with Albendazole and Praziquantel: A Retrospective Cohort Study. *Biomedicines*. 2026;14(8):1734. doi:10.3390/biomedicines14081734

9. Sitnikov I, Halil E, Atanasova N, Popova P, **Kostadinov K**, Ezeldin F, Judge E, Atanassova P. External Validation of BrainScan AI for Detection of Hemorrhagic Stroke on Non-Contrast Head CT. *The EuroBiotech Journal*. 2026;10(3):258–269. doi:10.2478/ebtj-2026-0019

10. **Kostadinov K**, Dzhambov AM, Burov A, Helbich M, Markevych I, Nieuwenhuijsen MJ, Dimitrova D. Evidence-Based Policy for Urban Environmental Health: A Cross-Sectional Stakeholder Survey in Bulgaria. *Urban Science*. 2026;10(6):312. doi:10.3390/urbansci10060312

11. Babacheva V, **Kostadinov K**, Goranova-Marinova V, Hristova M, Atanassova P. Cognitive Performance in Transfusion-Dependent Adults with β-Thalassemia in Bulgaria: A Case–Control Study. *Neurology International*. 2026;18(6):101. doi:10.3390/neurolint18060101

12. Halil E, Sitnikov I, Atanasova N, Popova P, **Kostadinov K**, Ezeldin F, Atanassova P. The Role of AI-Based Software BrainScan in the Interpretation of Non-Contrast Head CT in Acute Ischemic Stroke: An External Validation Study. *Neurology International*. 2026;18(6):100. doi:10.3390/neurolint18060100

13. **Kostadinov K**, Goodwin J, Groen G, Jörns-Presentati A, Heffernan S, O'Donovan Á, O'Malley M, Biskupovic Goulding R, O'Mahony J, Allen S, Curtin M, Haapalainen S, Aho J, Petrova G, Ivanova S, Lalova V, Korhonen J. Virtual Reality–Enhanced Training for Trauma-Informed Care Among Residential and Child Mental Health Professionals: Pre-Post Evaluation Study. *JMIR Medical Education*. 2026;12:e86543–e86543. doi:10.2196/86543

14. Petrov B, **Kostadinov K**, Dechev I, Eneva K. Etiological spectrum of sexually transmitted infections in infertile men: a monocentric retrospective study. *Folia Medica*. 2026;68(2):e172980. doi:10.3897/folmed.68.e172980

15. Rupcheva MY, **Kostadinov K**, Tashev R, Markova P, Zheleva V, Chterev M, Atanasova M, Petrov MM, Murdjeva M. Comparative Evaluation of Multiplex Real-Time PCR, Standard Urine Culture, and Rapid Nephelometric Screening in Patients with Complicated Urinary Tract Infections. *Diagnostics*. 2026;16(6):919. doi:10.3390/diagnostics16060919

16. Arabadzhiev Z, **Kostadinov K**, Stoyanova K, Velchev V, Cherneva M, Tomcheva S. Introduction to Evidence-Based Positive Psychotherapy. *The Global Psychotherapist*. 2026;6(1):28–38. doi:10.52982/lkj288

17. Shindova M, Katsarov P, Hristozova E, **Kostadinov KR**, Onov M, Belcheva A. Antimicrobial Activity of Root Canal Filling Materials for Endodontic Treatment in Primary Dentition. *Journal of IMAB – Annual Proceeding (Scientific Papers)*. 2026;32(1):6691–6697. doi:10.5272/jimab.2026321.6691

18. Bechev K, Yotova N, Kanarev M, Petrova A, **Kostadinov K**, Markov G, Markov D. Morphological Analysis of the Cavernous Segment of the Internal Carotid Artery: A Retrospective, Single Center Study of Its Clinical Significance. *Diagnostics*. 2025;15(23):3072. doi:10.3390/diagnostics15233072

19. Raycheva R, **Kostadinov K**, Rangelova V, Kevorkyan A. Economic Analyses of COVID-19 Interventions: A Narrative Review of Global Evidence. *Healthcare*. 2025;13(24):3249. doi:10.3390/healthcare13243249

20. **Kostadinov K**, Raycheva R, Zdravkova-Aneva I, Shopova M, Ovchinnikov E, Petkov P. Accuracy of Budget Impact Projections in Bulgarian Health Technology Assessment: A Five-Year Validation Study (2020–2025). *Healthcare*. 2025;13(22):2990. doi:10.3390/healthcare13222990

21. Rupcheva MY, **Kostadinov K**, Kalchev Y, Gardzheva P, Hristozova E, Rachkovska Z, Lengerova G, Angelova A, Murdjeva M, Petrov MM. Urinary Tract Infections in a Single-Center Bulgarian Hospital: Trends in Etiology, Antibiotic Resistance, and the Impact of the COVID-19 Pandemic (2017–2022). *Antibiotics*. 2025;14(10):982. doi:10.3390/antibiotics14100982

22. Stefanov R, Raycheva R, **Kostadinov K**, Stefanov G, Zdravkova-Aneva I, Dzhambazova E, Iskrov G. Access to Orphan Medicinal Products in Bulgaria: An Analysis of the Positive Drug List and Individual Access Schemes. *Healthcare*. 2025;13(18):2258. doi:10.3390/healthcare13182258

23. Halil E, **Kostadinov K**, Traykova N, Atanasova N, Atliev K, Dzhambazova E, Atanassova P. Novel Computed Tomography Perfusion and Laboratory Indices as Predictors of Long-Term Outcome and Survival in Acute Ischemic Stroke. *Neurology International*. 2025;17(9):136. doi:10.3390/neurolint17090136

24. Korhonen J, Lahti M, **Kostadinov K**, Erjavec K, Istomina N, Ivanova S, Lagiou A, Lalova V, Makutienė M, Notara V, Ollikkala H, Petrova G, Sakellari E, Sukyte D, Laaksonen C. Fostering Mental Health Literacy Among Primary School Professionals: Evaluating the Impact of an Online Training Program 'Well@School'. *International Journal of Environmental Research and Public Health*. 2025;22(3):435. doi:10.3390/ijerph22030435

25. **Kostadinov K**, Iskrov G, Musurlieva N, Stefanov R. 'It Felt Like Finding Hope Only to Lose It Again': A Grounded Theory Study of Rare Cancer Policies in Bulgaria. *Journal of Cancer Policy*. 2025:100570. doi:10.1016/j.jcpo.2025.100570

26. Dzhambazova E, **Kostadinov K**, Tsenkova-Toncheva L, Galabova F, Ezeldin F, Iskrov G, Stefanov R. The Socio-Economic Burden of Spinal Muscular Atrophy: A Cost-of-Illness Study in Bulgaria. *Healthcare*. 2025;13(4):401. doi:10.3390/healthcare13040401

27. **Kostadinov K**, Iskrov G, Musurlieva N, Stefanov R. An Evaluation of Rare Cancer Policies in Europe: A Survey Among Healthcare Providers. *Cancers*. 2025;17(2):164. doi:10.3390/cancers17020164

28. Groen G, Jörns-Presentati A, Goodwin J, Petrova G, O'Donovan Á, **Kostadinov K**, Haapalainen S, Korhonen J. Safe4Child: Trauma-informed approaches to support staff working with children and adolescents exhibiting behaviours that challenge. *Standpunkt: sozial*. 2025;35(1):1–13. doi:10.15460/spsoz.2025.35.1.240

29. Goodwin J, O'Malley M, O'Donovan A, Allen S, Curtin M, Goulding R, Groen G, Heffernan S, Ivanova S, Korhonen J, Jörns-Presentati A, **Kostadinov K**, Lalova V, O'Mahony J, Petrova G, Vainio V, Lahti M. Staff Experiences of Working with Children and Adolescents Engaging in Behaviours That Challenge in Mental Health and Paediatric Inpatient Environments: A Qualitative Exploratory Study across Four European Countries. *Child Care in Practice*. 2025;31(3):380–400. doi:10.1080/13575279.2024.2407085

30. Petrova G, **Kostadinov K**, Lalova V, Ivanova S. Formed Attitudes and Benefits of Training with Virtual Reality Glasses Among Nursing Students and Educators (in Bulgarian). *Management and Education (Управление и образование)*. 2025;21:5. link

31. Stoeva V, Batselova H, **Kostadinov K**, Atliev K. Assessment of Occupational Risk Exposures and the Degree of Implementation of Standard Protective Equipment among Medical Personnel. *Journal of IMAB - Annual Proceeding (Scientific Papers)*. 2025;31(4). doi:10.5272/jimab.2025314.6525

32. Bechev K, Yotova NI, **Kostadinov K**, Ilyov I, Markov D. Anatomical Features and Morphometric Characteristics of the Sphenoidal Sinus in MRI Studies. *Cureus*. 2024. doi:10.7759/cureus.66764

33. Raycheva R, **Kostadinov K**, Mitova E, Iskrov G, Stefanov G, Vakevainen M, Elomaa K, Man YS, Gross E, Zschüntzsch J, Röttger R, Stefanov R. Landscape analysis of available European data sources amenable for machine learning and recommendations on usability for rare diseases screening. *Orphanet Journal of Rare Diseases*. 2024;19(1):147. doi:10.1186/s13023-024-03162-5

34. Iskrov G, Raycheva R, **Kostadinov K**, Gillner S, Blankart CR, Gross ES, Gumus G, Mitova E, Stefanov S, Stefanov G, Stefanov R. Are the European reference networks for rare diseases ready to embrace machine learning? A mixed-methods study. *Orphanet Journal of Rare Diseases*. 2024;19(1):25. doi:10.1186/s13023-024-03047-7

35. **Kostadinov K**, Marinova Y, Dimitrov K, Hristova-Atanasova E, Iskrov G, Stefanov R. Navigating Gene Therapy Access: The Case of Bulgaria in the Context of the EU Regulatory Landscape. *Healthcare*. 2024;12(4):458. doi:10.3390/healthcare12040458

36. **Kostadinov K**, Popova-Sotirova I, Marinova Y, Musurlieva N, Iskrov G, Stefanov R. Availability and Access to Orphan Drugs for Rare Cancers in Bulgaria: Analysis of Delays and Public Expenditures. *Cancers*. 2024;16(8):1489. doi:10.3390/cancers16081489

37. Garnier N, Berghout J, Zygmunt A, Singh D, Huang KA, Kantz W, Blankart CR, Gillner S, Zhao J, Roettger R, Saier C, Kirschner J, Schenk J, Atkins L, Ryan N, Zarakowska K, Zschüntzsch J, Zuccolo M, Müllenborn M, Man YS, Goodman L, Trad M, Chalandon AS, Sansen S, Martinez-Fresno M, Badger S, Olden RWV, Rothmann R, Lehner P, Tschohl C, Baillon L, Gumus G, Gross E, Stefanov R, Iskrov G, Raycheva R, **Kostadinov K**, Mitova E, Einhorn M, Einhorn Y, Schepers J, Hübner M, Alves F, Iskandar R, Mayer R, Renieri A, Piperkova A, Gut I, Beltran S, Matthiesen ME, Poetz M, Hansson M, Trollmann R, Agolini E, Ottombrino S, Novelli A, Bertini E, Selvatici R, Farnè M, Fortunato F, Ferlini A. Genetic newborn screening and digital technologies: A project protocol based on a dual approach to shorten the rare diseases diagnostic path in Europe. *PLOS ONE*. 2023;18(11):e0293503. doi:10.1371/journal.pone.0293503

38. **Kostadinov K**, Musurlieva N. Historical Overview of the Definition of Rare Tumors in the Context of Health Policies. *Rare Diseases and Orphan Drugs Journal*. 2023;14(2):6–12. doi:10.36865/2023.v14i2.182

39. Guevara K, **Kostadinov K**. Neurological Soft Signs and Social Cognition in Patients with Schizophrenia: the Missing Link. *Proceedings of the Bulgarian Academy of Sciences*. 2023;76(9):1440–1448. doi:10.7546/CRABS.2023.09.15

40. Kelly P, Saab MM, Hurley EJ, Heffernan S, Goodwin J, Mulud ZA, O Malley M, O Mahony J, Curtin M, Groen G, Ivanova S, Jörns-presentati A, Korhonen J, **Kostadinov K**, Lahti M, Lalova V, Petrova G, O Donovan A. Trauma Informed Interventions to Reduce Seclusion, Restraint and Restrictive Practices Amongst Staff Caring for Children and Adolescents with Challenging Behaviours: A Systematic Review. *Journal of Child and Adolescent Trauma*. 2023. doi:10.1007/s40653-023-00524-2

41. Kiani F, **Kostadinov K**, Stefanov R, Mandova V. Dentophobia-latent Component Factor Analysis of Dental Concerns Assessment Scale. *Open Access Maced J Med Sci.* 2023;11(E):53–58. doi:10.3889/oamjms.2023.9749

42. **Kostadinov K**, Musurlieva N. Rare Tumors in the Context of Assisted Reproduction (in Bulgarian). *Journal "Medical Law and Healthcare" (Списание "Медицинско право и здравеопазване").* 2023;4:6–12.

43. Raycheva R, **Kostadinov K**, Mitova E, Bogoeva N, Iskrov G, Stefanov G, Stefanov R. Challenges in mapping European rare disease databases, relevant for ML-based screening technologies in terms of organizational, FAIR and legal principles: scoping review. *Frontiers in Public Health.* 2023;11. doi:10.3389/fpubh.2023.1214766

44. Ignatova V, **Kostadinov K**, Vassileva E, Muradyan N, Stefanov G, Iskrov G, Stefanov R. Socio-Economic Burden of Myasthenia Gravis: A Cost-of-Illness Study in Bulgaria. *Front. Public Health.* 2022;10. doi:10.3389/fpubh.2022.822909

45. Hubenova M, **Kostadinov K**. Change in eating habits during the COVID-19 confinement. *General Medicine.* 2022;24(1):23 – 29. link

46. **Kostadinov K**, Stefanov G, Iskrov G, Stefanov R. Epidemiology of Acquired Thrombotic Thrombocytopenic Purpura. *Rare Diseases and Orphan Drugs.* 2020;11(3):36–44. doi:10.36865/2020.v11i3.122

Commentaries and letters (1)

47. **Kostadinov K**. Selection bias and inferential validity in fever of unknown origin studies. Comment on: Poposki et al. Visceral leishmaniasis as a leading cause of fever of unknown origin in immunocompetent adults. *Folia Medica.* 2026;68(3):e184403. doi:10.3897/folmed.68.e184403

Books and monographs (1)

48. Kevorkyan A, **Kostadinov K**, Rangelova V, Raycheva R. COVID-19 in Bulgaria: Epidemiological Analysis (in Bulgarian). *Paisii Hilendarski University Press (Университетско издателство „Паисий Хилендарски").* 2023.

Book chapters (1)

49. Petrova G, **Kostadinov K**, Lalova V, Ivanova S. Virtual simulation education of students from specialty of "Nursing care". In: Chmiel Z, Sałacińska I, Lesiak A, editors. *Medical simulation and international training ground for teaching and scientific experiences.* Rzeszów University Press; 2026:91–100. link

Conference abstracts published in journals (9)

50. Dzhambazova E, **Kostadinov K**, Ezeldin F, Iskrov G. The socioeconomic burden of osteoporosis: a cost of illness study in Bulgaria. *European Journal of Public Health.* 2025;35(Supplement_4):ckaf161.1078. doi:10.1093/eurpub/ckaf161.1078

51. Dzhambazova EK, **Kostadinov KR**, Ezeldin F, Iskrov G. RWD235 The socio-economic burden of terminal cancer: real-world evidence from Bulgaria. *Value in Health Regional Issues.* 2025;49:101385. doi:10.1016/j.vhri.2025.101385

52. Dzhambazova EK, **Kostadinov KR**, Ezeldin F, Iskrov G. RWD73 The socio-economic burden of end-stage kidney disease: real-world evidence from Bulgaria. *Value in Health Regional Issues*. 2025;49:101303. doi:10.1016/j.vhri.2025.101303

53. Halil E, Atanasova P, Traykova N, **Kostadinov K**. CT Perfusion and Laboratory Markers as Predictors of Long-Term Functional Outcome and Survival in Acute Ischemic Stroke. *International Journal of Stroke*. 2025;20(2_suppl). doi:10.1177/17474930251371440

54. Velikova T, **Kostadinov K**, Naseva E, Rubinstein T, Sarkar M, Mangla L, Makowska J, Micheroli R, Ferreyra L, Kobert L, Parodis I, Nikiphorou E, Shinjo S, Caballero Uribe CV, Venerito V, Agarwal V, Gupta L. Social Determinants Shape Health Outcomes in Idiopathic Inflammatory Myopathies: Insights from Lived Experience of 1,252 Patients. *Arthritis & Rheumatology*. 2025;77(suppl 9). link

55. **Kostadinov K**, Hristova-Atanasova E, Musurlieva N, Iskrov G, Stefanov R. Mapping the Policy Alternatives for Rare Cancer. *European Journal of Public Health*. 2024;34:842-843. doi:10.1093/eurpub/ckae144.2172

56. Petrova G, **Kostadinov K**, Lalova V, Ivanova S. Exploring Empathy Dynamics in Virtual Patient Training. *European Journal of Public Health*. 2024;34(Supplement_3):444-445. doi:10.1093/eurpub/ckae144.1139

57. Raycheva R, **Kostadinov K**. HTA7 The Price of Innovation – Oncology Treatments Expenditures: Case from Bulgaria. *Value in Health*. 2023;26(6):S260. doi:10.1016/j.jval.2023.03.1436

58. Raycheva R, **Kostadinov K**. HTA69 Delay of Innovative Oncology Treatments - Case From Bulgaria. *Value in Health*. 2022;25(12):S309–S310. doi:10.1016/j.jval.2022.09.1529

Other scholarly outputs (conference proceedings) (2)

59. Markevych I, Baumbach C, Burov A, Helbich M, Dimitrova D, **Kostadinov K**, Nieuwenhuijsen MJ, Dzhambov AM. Land cover diversity and prevalence of allergic rhinitis/hay fever among Bulgarian urban adults. *ISEE 2026*. 2026.

60. Burov A, Dzhambov A, Dimitrova D, Markevych I, **Kostadinov K**, Helbich M, Nieuwenhuijsen M. Reintroduction of Health Impact Assessment in Bulgarian City Planning – Why and How? *AESOP Congress 2025*. 2025.

Research software and open science

`2024 - Present`

- `kkstatfun` — R package for medical statistics, epidemiology and health economics. A single tidy, pipe-friendly and `group_by()`-aware interface covering measures of association, standardisation, diagnostic accuracy, agreement, survival analysis, regression, causal inference and quasi-experimental designs, health technology assessment, and infectious disease modelling. Current release **v1.3.0**, 146 functions, MIT licence.
  - Source: github.com/kostadinoff/kkstatfun · Archive: doi:10.5281/zenodo.18936019 · Documentation: kkstatfun function reference

2025

- **PhD dissertation appendices and analysis code** — reproducible materials accompanying the doctoral thesis on health policies in rare tumours: github.com/kostadinoff/phd_appendices

Ongoing

- **Reproducible analytical workflows** — Quarto-based reproducible reporting for manuscripts, teaching materials and policy analyses; version-controlled research workflows in Git and GitHub; literate analysis pipelines in R with figures and tables regenerated from source data.
- **Open educational and scholarly resources** — kostadinoff.github.io: teaching slides, 15-minute learning modules, learning tasks and the *Epi News* research digest, all published openly and generated from source (see *Teaching materials and textbooks*).

2024 - 2025

- **OpenAQ Community Ambassador** — advocacy for open air quality data, data transparency and participatory environmental health research.

September 2024

- **Open Science certification** | NASA Transform to Open Science curriculum

Teaching experience

Undergraduate and postgraduate teaching at the Faculty of Public Health, Medical University of Plovdiv, since 2020, for students of medicine, dental medicine, pharmacy and nursing, in both Bulgarian and English. Postgraduate teaching reaches participants from a range of countries and continents.

Course	Level	Semester	Format	Language	Years
Biostatistics / Medical Statistics	Undergraduate	Winter	Lectures, practicals	BG · EN	2021/22 – 2025/26
Medical Ethics	Undergraduate	Winter	Lectures, practicals	BG · EN	2021/22 – 2025/26
Public Dental Health	Undergraduate (Dental Medicine)	Winter	Practicals	BG · EN	2021/22 – 2025/26
Clinical Trial Design for Non-Interventional Studies	Postgraduate	Winter	Lectures, practicals	EN	2023/24 – 2025/26
Social Medicine and Public Health	Undergraduate	Summer	Lectures, practicals	BG · EN	2021/22 – 2025/26

Additional teaching responsibilities

- Course coordination and curriculum development in biostatistics and public health teaching
- Design, administration and marking of examinations; continuous student assessment
- Development of open teaching materials, practical guides and reproducible statistical exercises (see *Research software and open science*)
- International teaching mobility under Erasmus+ (see *International teaching mobility*)

Semester record

2025/26 Winter: Biostatistics | Medical Ethics | Public Dental Health | Clinical Trial Design for Non-Interventional Studies · Summer: Social Medicine and Public Health

2024/25 Winter: Biostatistics | Medical Ethics | Public Dental Health | Clinical Trial Design for Non-Interventional Studies · Summer: Social Medicine and Public Health

2023/24 Winter: Biostatistics | Medical Ethics | Public Dental Health | Clinical Trial Design for Non-Interventional Studies · Summer: Social Medicine and Public Health

2022/23 Winter: Biostatistics | Medical Ethics | Public Dental Health · Summer: Social Medicine and Public Health

2021/22 Winter: Biostatistics | Medical Ethics | Public Dental Health · Summer: Social Medicine and Public Health

Teaching materials and textbooks

2025 - Present

- ***Medical Statistics: A Practical Guide*** | Bulgarian and English editions | Open-access practical guide for undergraduate medical statistics teaching, with worked examples and reproducible exercises

2025 - Present

- ***Medical Ethics and Deontology*** | Bulgarian and English editions | Open-access teaching text for undergraduate medical ethics

2025 - Present

- ***Social Medicine and Public Health*** | Semester book, English edition | Course text for the English-language undergraduate programme

2023

- Kevorkyan A, **Kostadinov K**, Rangelova V, Raycheva R. ***COVID-19 in Bulgaria: Epidemiological Analysis*** (in Bulgarian) | Paisii Hilendarski University Press | Recorded as a teaching aid by the Medical University of Plovdiv; also listed under *Publications — Books and monographs*

All teaching texts are written in Quarto and published openly at kostadinoff.github.io.

Supervision and mentorship

December 2025 - June 2026

- **Academic Mentor** | Project BG05SFPR001-3.002-0001 "From Higher Education to Employment" | Formal mentorship under a nationally funded programme; supervision of students in practical placements

2024 - Present

- **Student research supervision and co-supervision** | Medical University of Plovdiv | Supervision of student scientific projects in epidemiology, biostatistics and microbiology, covering study design, analysis and preparation of conference and journal outputs

Student outputs arising from supervision and mentorship

- Rupcheva M, **Kostadinov K**, Tashev R, et al. — accelerated antimicrobial susceptibility testing in urinary tract infections | **First place, Best Poster**, European Biotechnology Congress 2026, Varna
- Markova P, Spasova M, Yaneva A, **Kostadinov K** — optimisation of GFR estimation for chemotherapy dosing in paediatric oncology | **Best Presentation prize**, Science and Youth 2025
- Budak A, **Kostadinov K**, Tashev R, Christozova E — antimicrobial susceptibility of *Staphylococcus aureus* nasal isolates in preclinical medical students | **Best Presentation prize**, Science and Youth 2024
- Razheva E, **Kostadinov K**, Iskrov G, Miteva-Katrandzhieva T, Stefanov R — regional disparities in the availability of physical therapy and rehabilitation in Bulgaria | Science and Youth 2025

Editorial and peer-review service

Since 2025

- **Associate Editor** | *JMIRx Med* | JMIR Publications

Since 2025

- **Peer reviewer** for international peer-reviewed journals in public health, health policy and epidemiology

Invited lectures and keynotes

13–15 July 2026

- **Invited Speaker** | Air Quality Reporter Briefing | Breathe Cities | London, United Kingdom

10 July 2026

- **Invited Speaker** | Working meeting of the Ministry of Health, Ministry of Agriculture and Food, Bulgarian Food Safety Agency, Regional Health Inspectorates and Information Services JSC, 9–11 July 2026 | Organised by the Bulgarian Society for Innovative Medicine | *"The Role of the National Health Information System (NHIS) in Communicable Disease Surveillance: Future Perspectives"* | **Kostadinov K** | Session on improving surveillance of influenza and other viral respiratory pathogens by integrating existing surveillance systems under a One Health approach

10–13 October 2024

- **Invited Lecture** | XVIII National Congress of Cardiology | Plovdiv, Bulgaria | *"The Role of Air Pollution in Cardiovascular Disease Genesis"* | **Kostadinov K**

14 September 2023

- **Invited Webinar Speaker** | EU Health Technology Assessment Regulation — The Role and Involvement of Cancer Patients | *"Landscape of Bulgarian HTA Regulations"* | **Kostadinov K** | Belgium

9–11 March 2021

- **Plenary Lecture** | Science and Youth 2021 | *"Changes in Physical Activity During Epidemic Control Measures"* | **Kostadinov K**, Hubenova M, Mandova V

Conference contributions

Presenter roles are stated where documented. Entries marked **Co-author** were presented by another member of the author team.

International

2026

- **ISEE 2026** — International Society for Environmental Epidemiology | *"Land cover diversity and prevalence of allergic rhinitis/hay fever among Bulgarian urban adults"* | Markevych I, Baumbach C, Burov A, Helbich M, Dimitrova D, **Kostadinov K**, Nieuwenhuijsen MJ, Dzhambov AM | Poster presentation | Co-author

24-25 April 2026

- **European Biotechnology Congress 2026** | Varna, Bulgaria | *"Accelerated antimicrobial susceptibility testing in patients with urinary tract infections: laser light scattering versus disc diffusion"* | Rupcheva M, **Kostadinov K**, Tashev R, Markova P, Zheleva V, Tsvetkova Y, Atanasova M, Petrov M | Poster presentation | Co-author | **First place, Best Poster**

11-14 November 2025

- **18th European Public Health Conference 2025** | Helsinki, Finland | *"The Socioeconomic Burden of Osteoporosis: A Cost of Illness Study in Bulgaria"* | Iskrov G, **Kostadinov K**, Dzhambazova E, Ezeldin F | E-poster presentation | Co-author

27 October 2025

- **ACR Convergence 2025** | Chicago, IL, USA | *"Social Determinants Shape Health Outcomes in Idiopathic Inflammatory Myopathies: Insights from Lived Experience of 1,252 Patients"* | Velikova T, **Kostadinov K**, Naseva E, et al. | Poster presentation | Co-author

28-30 September 2025

- **ISPOR Real-World Evidence Summit 2025** | Tokyo, Japan
  1. *"RWD235 The socio-economic burden of terminal cancer: real-world evidence from Bulgaria"* | Iskrov G, **Kostadinov K**, Dzhambazova E, Ezeldin F | Poster presentation | Co-author
  2. *"RWD73 The socio-economic burden of end-stage kidney disease: real-world evidence from Bulgaria"* | Iskrov G, **Kostadinov K**, Dzhambazova E, Ezeldin F | Poster presentation | Co-author

23-27 September 2025

- **56th Annual Meeting of the European Society for Paediatric Nephrology** | Greece | *"Optimization of eGFR Assessment in Pediatric Oncology: Limitations of the Schwartz Formula and Alternative Approaches"* | **Kostadinov K**, Yaneva A, Spasova M, Spasov N, Markova P | Poster presentation | Presenter

18-19 September 2025

- **XVII International Symposium "Challenges Affecting the Strengthening of the Role of Nurses in Modern Society"** | České Budějovice, Czech Republic | *"Developing Competencies for Specialized Healthcare for Patients with Psychological Trauma"* | **Kostadinov K**, Petrova G, Lalova V, Ivanova S | Oral presentation | Presenter

7-11 July 2025

- **AESOP Congress 2025** | Yıldız Technical University, Istanbul, Türkiye | *"Reintroduction of Health Impact Assessment in Bulgarian City Planning – Why and How?"* | Burov A, Dzhambov A, Dimitrova D, Markevych I, **Kostadinov K**, Helbich M, Nieuwenhuijsen M | Co-author

19-20 June 2025

- **International Scientific Conference "Education, Science, Economics and Technology"** | Burgas, Bulgaria | *"Formed attitudes and benefits of training with virtual reality glasses among nursing students and educators"* | Petrova G, **Kostadinov K**, Lalova V, Ivanova S | Co-author

12-15 November 2024

- **17th European Public Health Conference 2024** | Lisbon, Portugal | *DV.24 — Mapping the Policy Alternatives for Rare Cancer* | **Kostadinov K**, Hristova-Atanasova E, Musurlieva N, Iskrov G, Stefanov R | Poster presentation | Presenter

20-24 June 2024

- **Xth International Conference of Young Scientists** | Plovdiv, Bulgaria | *"Regulatory Framework for Market Authorization of Gene Therapies in the EU and the USA"* | Stefanov R, Iskrov G, Hristova-Atanasova E, **Kostadinov K**, Dimitrov K | Oral presentation | Co-author

7-10 May 2023

- **ISPOR 2023** | Boston, MA, USA | *"HTA7 The Price of Innovation — Oncology Treatments Expenditures: Case from Bulgaria"* | Raycheva R, **Kostadinov K** | Co-author

15-18 April 2023

- **33rd European Congress of Clinical Microbiology and Infectious Diseases (ECCMID)** | Copenhagen, Denmark | *"Impact of COVID-19 Vaccines – Data from Bulgaria"* | Kevorkyan A, **Kostadinov K**, Rangelova V, Raycheva R, Kunchev A | Co-author

4-11 March 2023

- **41st European Winter Conference on Brain Research (EWCBR)** | *"Immunological Reactivity Under Acute and Chronic Stress. Where Are We? Experience in Bulgaria"* | Ivanovska M, Kalfova T, Gardjeva P, **Kostadinov K**, Murdjeva M | Co-author

6-9 November 2022

- **ISPOR Europe 2022** | Vienna, Austria | *"HTA69 Delay of Innovative Oncology Treatments — Case From Bulgaria"* | Raycheva R, **Kostadinov K** | Co-author

20-23 October 2022

- **Joint Forum: 12th South-East European Conference and 32nd Annual Assembly of IMAB** | *"Antimicrobial Activity of Root Canal Filling Materials for Endodontic Treatment in Primary Dentition"* | Shindova M, Hristozova E, Katsarov P, Onov M, **Kostadinov K**, Toplev V, Belcheva A | Lecture | Co-author

National

28 October 2026

- **Round Table "Healthcare Expenditure in Bulgaria – Statistical Aspects"** | D. A. Tsenov Academy of Economics, Svishtov (hybrid) | *"Accuracy of Budget Impact Projections in Bulgarian Health Technology Assessment, 2020–2025"* | **Kostadinov K**, Raycheva R, Zdravkova-Aneva I, Shopova M, Ovchinnikov E, Petkov P | Oral presentation | Presenter

26 September 2026

- **8th National Conference of Epidemiology "One Health in a Dynamic Global World"** | Stara Zagora | *"Epidemiological Burden and Cost-Effectiveness of Universal Varicella Vaccination in Bulgaria: An Analysis of National Administrative Data with Dynamic Modelling"* | **Kostadinov K**, Raycheva R, Kevorkyan A, Rangelova V, Parmakova K | Oral presentation | Presenter

17-18 April 2026

- **Science and Youth 2026** | Plovdiv | *"Application of the Evidence-Based Psychotherapy Focusing on the Correlation Between Early Dysfunctional Schemas and Current Capabilities"* | Velchev V, Cherneva M, **Kostadinov K**, Stoyanova K, Arabadzhiev Z | Co-author

12-13 September 2025

- **XVI National Conference on Rare Diseases and Orphan Drugs** | *"Orphan Drugs in Bulgaria: The Path from Marketing Authorization to the Patient"* | Stefanov R, Iskrov G, Raycheva R, **Kostadinov K** | Oral presentation | Co-author

29 May - 1 June 2025

- **XII National Scientific Conference "HIV/AIDS and Exotic Parasitic and Infectious Diseases"** | *"Assessment of Occupational Risk Exposures and Behavior Among Medical Staff"* | **Kostadinov K**, Stoeva V, Batselova H, Atliev K | Oral presentation | Presenter

9-11 May 2025

- **Science and Youth 2025** | Plovdiv
  1. *"Regional disparities and dynamics in the availability of physical therapy and rehabilitation in Bulgaria"* | Razheva E, **Kostadinov K**, Iskrov G, Miteva-Katrandzhieva T, Stefanov R | Poster presentation | Co-author
  2. *"A rapid review of health outcomes associated with the food environment in cities"* | **Kostadinov K**, Dimitrova D, Helbich M, Burov A, Dzhambov A | Poster presentation | Presenter
  3. *"Optimization of GFR estimation for chemotherapy dosing in pediatric oncology"* | Markova P, Spasova M, Yaneva A, **Kostadinov K** | Oral presentation | Co-author | **Best Presentation prize**

10-11 October 2024

- **Scientific Conference "Environment and Health: Socio-Technical Barriers and Perspectives to Improve Quality of Life for Human Communities"** | Plovdiv University "Paisii Hilendarski", Plovdiv | *"Research on the Link Between Air Pollution, Urban Environment Characteristics and Population Health"* | Dzhambov A, **Kostadinov K**, Dimitrova D | Co-author

27-29 September 2024

- **Sixth National Epidemiology Conference "Challenges in Infectious Disease Prevention"** | *"Assessment of Occupational Risk Exposures and Compliance with Standard Precautions among Medical Staff"* | Stoeva V, Batselova H, **Kostadinov K**, Atliev K | Co-author

27-28 September 2024

- **Seventh Scientific Conference with International Participation "Public Health: A Look Toward the Future"** | *"Enhancing Patient-Centered Care through Simulation Training with Virtual Reality Glasses"* | Petrova G, **Kostadinov K**, Lalova V, Ivanova S | Co-author | **Best Presentation prize**

19-21 April 2024

- **Science and Youth 2024** | Plovdiv | *"Assessment of Antimicrobial Susceptibility of Staphylococcus aureus Nasal Isolates from Preclinical Medical Students at the Medical University of Plovdiv"* | Budak A, **Kostadinov K**, Tashev R, Christozova E | Co-author | **Best Presentation prize**

30 September - 1 October 2022

- **Fourth National Conference on Epidemiology "Infectious Diseases in Bulgaria: Challenges and Perspectives"** | *"COVID-19 in Bulgaria and the Impact of Vaccines on Hospitalizations, Mortality and Lethality"* | Kevorkyan A, **Kostadinov K**, Rangelova V, Raycheva R, Kunchev A, Serbezova A | Co-author

15-16 September 2022

- **XX Jubilee National Congress on Clinical Microbiology and Infections** | Bulgarian Association of Microbiologists
  1. *"Study of Urethral Microbiome with Androflor Screen in Men with Non-Specific Genital Symptoms During the COVID-19 Pandemic"* | Christozova E, Dermendzhiev T, Rachkovska Z, Georgieva V, **Kostadinov K**, Pavlov C, Murdjeva M | Published in proceedings | Co-author
  2. *"Omicron — Relief or Challenge?"* | Atanasova M, Korsun N, Komitova R, **Kostadinov K**, Aleksiev I, Raycheva R, Ivanov I, Glomb L, Petkova C, Djoglova L | Plenary lecture | Published in proceedings | Co-author

14-16 September 2021

- **XVIII National Congress on Clinical Microbiology and Infections** | Bulgarian Association of Microbiologists | *"Rapid Molecular Screening for Vaginal Candidiasis in Symptomatic Women"* | Christozova E, Rachkovska Z, Georgieva V, Dermendzhiev T, **Kostadinov K**, Vlahova M, Uchikova E, Murdjeva M | Poster presentation | Co-author

13-14 May 2021

- **XIV National Scientific-Technical Conference with International Participation "Ecology and Health"** | *"Modern Microbiological and Molecular Screening of Genital Infections in Symptomatic Non-Pregnant Women"* | Hristozova E, Rachkovska Z, Dermendzhiev T, Murdjeva M, Georgieva V, Uchikova E, **Kostadinov K** | Co-author

9–11 March 2021

- **Science and Youth 2021** | Plovdiv
  1. *"Changes in Dietary Habits and Behavior During COVID-19 Epidemic Control Measures"* | Hubenova M, **Kostadinov K**, Mandova V | Plenary lecture | Co-author
  2. *"Changes in Physical Activity During Epidemic Control Measures"* | **Kostadinov K**, Hubenova M, Mandova V | Plenary lecture | Presenter *(see Invited lectures and keynotes)*
  3. *"Oral Health Self-Assessment among Hemophilia Families"* | Mandova V, **Kostadinov K**, Stefanov R | Co-author

28 November 2020

- **Virtual Congress "Rare Diseases and Orphan Drugs"** | *"Acquired Thrombotic Thrombocytopenic Purpura along the Path of Challenges"* | **Kostadinov K** | Presenter

27–29 September 2019

- **11th Scientific Meeting of SOIBOM** | *"The Pathway of Patients with Cardiovascular Disease"* and *"Clinical Case of Valvular Atrial Fibrillation"* | **Kostadinov K** | Oral presentation | Presenter

Awards and honours

2026

- **First place, Best Poster** | European Biotechnology Congress 2026, Varna | Team award for accelerated antimicrobial susceptibility testing in urinary tract infections (Rupcheva M, **Kostadinov K**, Tashev R, et al.), funded under PSNIIR-MUP Contract BG-RRP-2.004-0007-C03

2025

- **Best Presentation prize** | Science and Youth 2025, Plovdiv | Optimisation of GFR estimation for chemotherapy dosing in paediatric oncology

2024

- **Best Presentation prize** | Seventh Scientific Conference "Public Health: A Look Toward the Future" | Simulation training with virtual reality glasses
- **Best Presentation prize** | Science and Youth 2024, Plovdiv | Antimicrobial susceptibility of *Staphylococcus aureus* nasal isolates
- **Best Presentation prize** | Open Medical Institute seminar, Public Health Strategy II | Salzburg, Austria
- **Competitive award** | National Science Programme "Young Scientists and Postdoctoral Researchers – 2" | Ministry of Education and Science
- **Competitive selection** | OpenAQ Community Ambassador Programme

2023

- **Best Presentation prize** | Open Medical Institute seminar, Public Health Strategy I | Salzburg, Austria

Policy, public health and knowledge-translation activities

2026

- **Commissioned analysis for the World Health Organization** — epidemiological and economic analysis of varicella and varicella vaccination in Bulgaria, supporting the introduction of varicella vaccine into the national immunisation programme

July 2026

- **Invited speaker**, interinstitutional working meeting of the Ministry of Health, Ministry of Agriculture and Food, Bulgarian Food Safety Agency, Regional Health Inspectorates and Information Services JSC, organised by the Bulgarian Society for Innovative Medicine — on the role of the National Health Information System in communicable disease surveillance, contributing to the revision of the national methodological guidance on influenza and acute respiratory infections
- **Invited speaker**, Breathe Cities Air Quality Reporter Briefing, London — translation of air quality evidence for media and city stakeholders

2025 - Present

- **Evidence for urban environmental policy** — stakeholder survey informing evidence-based policy for urban environmental health in Bulgaria (*Urban Science*, 2026); framework for reintroducing Health Impact Assessment into Bulgarian spatial planning (AESOP 2025)

September 2025

- **Expert contract, "Air for Health" Association** | Project 001491 "Engaging health professionals in the design of policies to combat air pollution in Sofia", 2024–2026 | Bulgarian translation and language editing of the World Health Organization guidance on air pollution for health professionals (104 pages)

2024 - 2025

- **OpenAQ Community Ambassador** — advocacy for air quality data transparency, open data use and community air pollution monitoring

2023 - Present

- **Health technology assessment and access to medicines** — analyses of orphan medicinal product access, HTA timeliness and budget impact accuracy informing national debate on reimbursement policy; briefing to EU-level cancer patient organisations on the EU HTA Regulation (2023)

2021 - Present

- **"Air for Health" Physician Network** — public engagement and professional advocacy on air pollution and health

International teaching mobility

19-23 January 2026

- **Erasmus+ Staff Mobility for Teaching** | Tartu Health Care College, Estonia | 16 teaching hours | Subjects taught: epidemiological methods and health policy analysis

4-8 September 2023

- **Erasmus+ Staff Mobility for Teaching** | University of West Attica, Athens, Greece | Subjects taught: epidemiological methods and health policy analysis

International professional development — research and training visits to Ireland, Greece, Austria, Estonia, Germany, Finland and the United Kingdom.

Professional service and memberships

Elected offices and service

2024 - 2026

- **Board Member** | Youth Scientific Society "Asclepius"

2025 - Present

- **Associate Editor** | *JMIRx Med*, JMIR Publications (see *Editorial and peer-review service*)

Professional memberships — international

Since 2026 Balkan Association for Vector-Borne Diseases (BAVBD) | Member, Epidemiology and Biostatistics Task Force

Since 2026 American College of Epidemiology (ACE) | Member (MACE)

Since 2026 TEPHIConnect

Since 2025 International Epidemiological Association (IEA)

Since 2022 World Federation of Public Health Associations (WFPHA)

Since 2020 European Public Health Association (EUPHA)

Professional memberships — national

Since 2020 Bulgarian Scientific Society for Public Health

Since 2020 Association for Social Research and Applied Research Practices

Since 2017 Bulgarian Medical Association

Since 2017 Bulgarian Society of Cardiology

Since 2016 Youth Scientific Society "Asclepius"

Since 2016 Association of Trainers and Researchers in General Medicine in Bulgaria

Non-governmental organisations

2021 - Present

- **"Air for Health" Physician Network**

2024 - 2025

- **OpenAQ Community Ambassador** | OpenAQ

Earlier leadership

2012 - 2017

- **Bulgarian Medical Students' Association** | Chair of the Supervisory Board 2015–2016 | Secretary 2017

2011 - 2017

- **Student Council, Medical University of Plovdiv** | President 2014 | President of the Student Assembly 2015 | Chair of the General Assembly 2013–2015

Selected advanced training and continuing professional development

Selected advanced training (2020 – present)

September 2026 - present

- Course | Alternative Economic Models | Circular Economy Institute | Online | 7-week training (selected as one of 20 participants)

9-11 September 2026

- Thematic course | Vaccinology | Medical University of Pleven | Online

July 2026

- GenEpi-BioTrain virtual training 28 | Genomics and epidemiology of polioviruses: how a virus resists eradication? | ECDC | Online | 17 and 20 July 2026 | Certificate of completion (code sVGYAN5bMP)
- E-learning course | Community engagement and socially vulnerable populations | ECDC | 4 July 2026 | Certificate of completion (code V2swzV62FB, 2 APHEA credits)

March 2026

- E-learning course | Behavioural and social research methodologies for the prevention and control of infectious diseases | ECDC | 19 March 2026 | Certificate of completion (code aDBwAbupxJ, 2 APHEA credits)
- E-learning course | Multisectoral collaboration for the implementation of a One Health approach to communicable diseases | ECDC | Certificate of completion

- GenEpi-BioTrain virtual training 24 | Antigen surveillance: from evolution to immune escape | ECDC | 23 and 25 March 2026 | Certificate of completion
- AURORAE webinar | Serology of seasonal influenza | ECDC/AURORAE | 12 March 2026 | Certificate of completion

January 2026

- GenEpi-BioTrain virtual training 22 | Genomic epidemiology and outbreak investigation of *Brucella* spp. in a One Health context | ECDC | Certificate of completion
- AURORAE webinar | INSaFLU-TELEVIR bioinformatics platform: virus metagenomic detection | ECDC/AURORAE | Certificate of completion

October 2025 - January 2026

- **InfektEpiLehre** | Robert Koch Institute (RKI), Department of Infectious Disease Epidemiology | Curriculum for specialised infectious disease epidemiology | Berlin, Germany

November 2025

- Training | Epidemiology and Econometrics with Stata | Sabiha Gulshan | Certificate of completion

September 2025

- GenEpi-BioTrain virtual training 20 | *Klebsiella pneumoniae* genomic epidemiology and antimicrobial resistance | ECDC
- Training | Stata Basic, Intermediate and Advanced — data handling to advanced analysis | Sabiha Gulshan | Certificate of completion

August 2025

- Micro-learning course | Assess and mitigate the risk of locally acquired *Aedes*-borne viral diseases in the EU/EEA | ECDC

July 2025

- Micro-learning course | Investigate an outbreak | ECDC

June 2025

- GenEpi-BioTrain virtual training 18 | Empowering AMR research through R: analysis and visualisation | ECDC

April 2025

- Micro-learning course | Role of social and behavioural sciences in the prevention and control of infectious diseases | ECDC

17-20 February 2025

- GenEpi-BioTrain virtual training 14 | R data analysis and visualisation | ECDC

September 2024 - 2026

- Professional training | SQL: Data Analysis and Reporting | 60 hours | Certificate of completion

- Professional training | Python: Data Analysis and Machine Learning | 75 hours | Certificate of completion

September 2024

- Course | Open Science | NASA Transform to Open Science | Certificate of completion
- Micro-learning course | Evaluating completeness of study reporting | ECDC

August 2024

- Micro-learning course | Evaluating the study question | ECDC

26 May - 1 June 2024

- Certificate course | Open Medical Institute seminar | Public Health Strategy II | Maastricht University Programme | Salzburg, Austria | **Best Presentation prize**

3-9 December 2023

- Certificate course | Open Medical Institute seminar | Public Health Strategy I | Maastricht University Programme | Salzburg, Austria | **Best Presentation prize**

January - March 2023

- Certificate course | Introduction to Bayesian Data Analysis | Prof. Dr. Shravan Vasishth and Dr. Anna Laurinavichyute | University of Potsdam, Germany | openHPI Programme | Certificate of completion

10-15 October 2022

- Certificate course | Open Medical Institute seminar | Economic Evaluation in Healthcare | Maastricht University Programme | Salzburg, Austria

September - December 2021

- Course | Machine Learning | SoftUni | Sofia, Bulgaria

June 2021

- Intensive specialised course | Academic Writing and Systematic Reviews | Contract 06-DK1/6 of 29 March 2021 | "COVID-19 HUB — Information, Innovations and Implementation of Integrative Scientific Developments" | Ministry of Education and Science

22-26 March 2021

- Course | Biostatistics with IBM SPSS Statistics | PhD School | Medical University of Plovdiv

March 2020

- English for Academic Purposes | Irina Mitrcheva, PhD | Department of Language Training | Medical University of Plovdiv
- Language Issues in Scientific and Medical Text | Assoc. Prof. Milieva, PhD | Department of Language Training | Medical University of Plovdiv

February 2020

- Postgraduate training | Promoters of Advanced Oncogenetics | HOPE — How Oncogenetics Predicts and Educates | Erasmus+ Programme 2018-1-RO01-KA202-049189

Earlier training record (2011 – 2019)

24-25 October 2019

- Course | Third School of Echocardiography | New echocardiographic methods in cardiology | Sofia, Bulgaria

22-23 February 2019

- Course | Second School of Echocardiography | Echocardiographic assessment of congenital and acquired cardiovascular diseases | Sofia, Bulgaria

19-20 October 2018

- Course | First School of Echocardiography | Introduction to echocardiographic diagnosis | Sofia, Bulgaria

1-2 December 2017

- Course | Critical Appraisal and Evidence-Based Medicine | Narinder Gosall and Gurpal Gosall | Sofia, Bulgaria

10 November 2017

- Course | Echocardiographic assessment of cardiac function | National Cardiology Hospital | Sofia, Bulgaria

24-28 April 2017

- Training | Medical support in radiation and nuclear accidents and terrorism emergency preparedness | National Institute of Radiobiology and Radiation Protection | Sofia, Bulgaria

20-21 October 2014

- Training | Diagnosis, care and treatment of people living with HIV/AIDS | Global Fund to Fight AIDS, Tuberculosis and Malaria | Plovdiv, Bulgaria

16 December 2012

- Training | Basic surgical skills | Tokuda Hospital Sofia | Sofia, Bulgaria

22-23 November 2012

- Training | National Programme for HIV and Sexually Transmitted Infections Prevention and Control 2008–2015 | Treatment and care for people living with HIV/AIDS and reducing stigma and discrimination | Plovdiv, Bulgaria

24-25 September 2011

- Training | Emergency care in medicine | Medical University of Plovdiv

Research methods expertise

- **Observational epidemiology** — cohort, case–control and cross-sectional designs; retrospective analysis of routinely collected administrative health data
- **Regression modelling** — linear, logistic, ordinal, Poisson and modified-Poisson models; model specification, validation and calibration
- **Survival analysis** — Kaplan–Meier estimation, Cox proportional hazards models, restricted mean survival time and competing risks
- **Longitudinal and time-series methods** — trend analysis, interrupted time series and surveillance analytics
- **Health-economic evaluation** — cost-of-illness, cost-effectiveness and budget-impact analysis; probabilistic sensitivity analysis
- **Real-world evidence** — analysis of payer, registry and hospital administrative datasets
- **Diagnostic accuracy and prediction** — sensitivity and specificity, ROC and AUC analysis, external validation, agreement and reliability
- **Evidence synthesis** — systematic, scoping and rapid reviews
- **Bayesian analysis** — Bayesian estimation with informative and sceptical priors
- **Reproducible statistical programming** — literate, version-controlled analysis pipelines and open research software

Technical skills

`Programming and statistical computing`

- R · Python · SQL · R package development

`Statistical software`

- Stata · SPSS · SAS · Jamovi · JASP

`Reproducible research`

- Quarto · Markdown · Pandoc · LaTeX

`Development and version control`

- Git · GitHub · VS Code · RStudio

`Systems and databases`

- Linux / WSL · Bash · MySQL

`Web`

- HTML · CSS

Languages

Bulgarian

- Native speaker

English

- First Certificate in English (FCE), Cambridge English Language Assessment | Reading C1 · Writing B2 · Use of English C1 · Listening B2 · Speaking C1
- Working language for teaching, research and publication